

Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques

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Abstract -The goal of this study is to use machine learning techniques to anticipate retail vendors' adoption of digital payment platforms. Data was collected from 270 merchants throughout 10 major retail markets in Bangalore, with a focus on small and medium-sized businesses. Using multiple regression, exploratory factor analysis, and support vector machines (SVM), the study identifies significant factors driving adoption, including perceived usefulness, social influence, and compatibility. The data show that, while perceived usefulness and social influence are important drivers, other factors such as perceived ease of use and technical support have limited effect. The SVM model was moderately accurate in predicting platform adoption. The report also evaluates customer satisfaction across various digital payment platforms and predicts future trends using ARIMA models. These findings have significant implications for improving digital payment adoption tactics among retailers.

Keywords- Digital payment adoption, retail vendors, machine learning, exploratory factor analysis, support vector machine, ARIMA.

I. INTRODUCTION

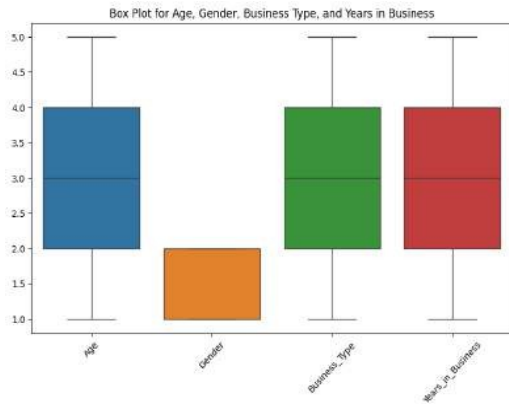
The rapid expansion of digital technologies has fundamentally altered the retail scene, particularly in terms of payment options. Traditional cash-based payments are slowly giving way to the convenience, speed, and security of digital payment platforms [1]–[3]. However, usage of these platforms by retail vendors can differ significantly. Knowing these factors is of utmost importance as relevant knowledge will benefit policymakers, platform providers, and retailers [4]–[6]. Verifiably, PhonePe, Paytm, GooglePay, and AmazonPay are some of the most used digital payment platforms in India [7], [8]. Despite the convenience that these digital payment platforms present, ongoing usage by small and medium-sized retail businesses is not fully established. Elements that contribute to the adoption process include perceived ease of use, perceived usefulness, social influence, and compatibility [1], [9]. The characteristics mentioned above will be studied in this project, and machine learning tools will be used to predict the likelihood of adoption by retail vendors.

Located in a bustling metropolitan area with an extensive retail landscape, Bangalore presents an ideal setting for our study. The emphasis on Bangalore's retail sector, which includes both an organized and unorganized sector, yields a spectrum of vendor types and vendor behaviors [4], [10]. By studying a particular market segment, one can extract these insights and apply them generically to other similar metropolitan markets in India. The study employed a robust approach that includes, for example, data collection via structured questionnaires, statistical analysis using multiple regression and exploratory factor analysis (EFA), as well as prediction using support vector machines (SVM), but also utilized ARIMA models to forecast future trends of digital payment adoption from Google Trends historical data [11]–[13]. The results of the study could inform strategy and improve practice to support the increased adoption of digital payments. Helpful to practitioners, the study identified the drivers of adoption, thus providing stakeholders practical insights to make decisions about the products they promote around payments and their services. The ultimate aim is to ease the transition to digital payments, improving retail vendors' efficiency and competitiveness.

II. LITERATURE REVIEW

In recent years, a significant amount of research has focused on the use of digital payment platforms because it is important to understand and improve the digital shift of retail. Several studies have examined the adoption process, and important factors were identified to drive retail stores decisions to accept digital payment options such as Technology Acceptance Model, perceived ease of use and perceived usefulness.[13] Perceived ease of use can be defined as the level to which an individual believes that using a specific system will require no or very little effort.[14] Perceived usefulness is defined as the 'level to which an individual believes that using the system will enhance their performance.'[14] This was an explanation of the original TAM framework which grows to consider social influence and facilitating conditions. Social Influence which is the 'degree to which an individual perceives that important others believe he or she should use the new systems'[15] was identified as an important factor as well. The Unified Theory of Acceptance and Use of Technology (UTAUT) model identifies social influence as an important predictor of behavioral intention to use [16].

Several studies have examined how retailers are adopting digital payment platforms. Perceived security and trust have been found to be important factors in retailers' adoption of mobile payment options [5]. This is consistent with [9], which also emphasizes the role of trust in adopting technology in situations involving financial-related transactions. The role of compatibility – how well an innovation fits into the potential adopter's existing values, prior experiences, and current needs – is also considered to play an important role. Compatibility has been examined in internet banking adoption and could be applied to digital payment platforms. Researching to predict technology adoption using machine learning techniques has become increasingly popular



[7], [17]. Machine learning techniques, such as support vector machines (SVM), have been used to explore complex datasets and uncover patterns that traditional statistics may have missed. It is found that using machine learning can allow for better prediction of technology adoption compared to traditional techniques.

A. Research Gap

While significant research has been conducted on the factors that influence the adoption of digital payment platforms, there is a noticeable gap in studies that integrate these elements with machine learning approaches to predict acceptance among retail vendors. Most previous research use traditional statistical approaches, which may not capture the complex relationships between variables. Hence, the present study tried to attain the below objectives for this study.

B. Research Objectives

1. Identify and assess the primary factors that influence retail vendors' adoption of digital payment platforms.
2. Using machine learning techniques, forecast the likelihood of adoption based on these factors.
3. To assess customer satisfaction across various digital payment platforms and estimate future adoption patterns.

C. Methodology

Bangalore city has 10 major retail markets specifically for small and medium vendors, both organized and unorganized. The entire city might have more than 30,000 merchants as the population for the study[18], [19]. Based on a scientific sampling procedure using a 90% confidence level, a 5% margin of error, and a 50% population proportion, a sample size of 270 was decided for the study[20]. To collect data from the 10 major retail markets in Bangalore for the sample of 270 merchants, an equal allocation stratified random sampling method have been used. Each market will

contribute 27 merchants to the sample, ensuring coverage across all markets. Merchants been randomly selected from

Dep. Variable:	Business_Type_2	R-squared:	0.003
Model:	OLS	Adj. R-squared:	-0.009
Method:	Least Squares	F-statistic:	0.2356
Date:	Fri, 19 Jul 2024	Prob (F-statistic):	0.871
Time:	02:46:49	Log-Likelihood:	-135.36
No. Observations:	270	AIC:	278.7
Df Residuals:	266	BIC:	293.1
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025
const	0.2447	0.092	2.667	0.008	0.064
Digital_Platforms	-0.0014	0.018	-0.077	0.939	-0.036
Number_of_Transactions	-0.0145	0.017	-0.836	0.404	-0.049
Day_Transaction	0.0007	0.017	0.040	0.968	-0.032

Omnibus:	61.599	Durbin-Watson:	1.864
Prob(Omnibus):	0.090	Jarque-Bera (JB):	101.142
Skew:	1.494	Prob(JB):	1.09e-22
Kurtosis:	3.249	Cond. No.	21.0

both organized and unorganized sectors within each market. Data have been gathered through structured questionnaires administered in person. All participants provided informed consent, and the obtained data was anonymised to preserve the respondents' privacy. The study followed ethical requirements for conducting human subjects' research.

III. ANALYSIS AND INTERPRETATION

A. Descriptive Statistics

Descriptive Statistics of Retail Vendors provides a comprehensive overview of vendor demographics, including age, gender, business type, and years in business, by summarizing key measures such as mean, median, mode, and standard deviation.

TABLE 1: DESCRIPTIVE STATISTICS OF RETAIL VENDORS

Variable	mean	std	min	25%	50%	75%	max
Age	3.00	1.446	1	2	3	4	5
Gender	1.55	0.497	1	1	2	2	2
Business Type	3.13	1.397	1	2	3	4	5
Years in Business	3.15	1.437	1	2	3	4	5

Fig.1 Box Plot

The clear insights uncover the taking after bits of knowledge into the merchant demographics: The normal age of sellers is roughly 3.00, with a standard deviation of 1.45, showing inconstancy within the age run. Gender dissemination is near to adjusted, with a cruel esteem of 1.56, proposing a prevalence of one sex. The cruel for trade sort and a long time in commerce are both around 3.13 and 3.16, individually, with comparable standard deviations of around 1.40, reflecting differing qualities in trade sorts and involvement levels among merchants. In general, these measurements outline a assorted test of merchants in terms of age, gender, business type, and experience.

B. Digital Payment Platform Usage Patterns

Digital Payment Platform Usage Patterns explores the relationship between business type and various usage metrics such as digital platforms, number of transactions per day, and transaction amounts, using multiple regression analysis to identify key factors influencing business type preferences. The objective is to determine how different digital payment platform usage patterns affect the classification of business types.

TABLE 2: DIGITAL PAYMENT PLATFORM USAGE PATTERNS

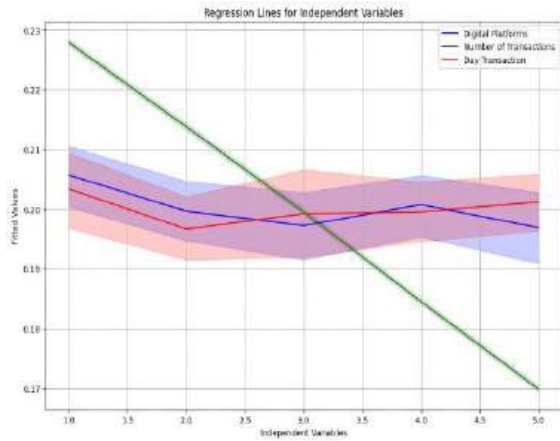


FIG.2 REGRESSION LINES

The analysis in Table 2 assesses how various digital payment platform usage patterns influence the classification of business types using multiple regression. The results indicate a very low R-squared value of 0.003, suggesting that the model explains almost none of the variability in business type based on the included usage metrics. The F-statistic is not statistically significant ($p = 0.871$), reinforcing the conclusion that the digital platforms, number of transactions, and daily transaction amounts have minimal impact on differentiating business types. Thus, the factors examined do not significantly affect the classification of business types in this context.

C. Factors Influencing Adoption

Factors Influencing Adoption identifies key factors affecting the adoption of digital payment platforms using Exploratory Factor Analysis (EFA) on variables such as perceived ease of use, perceived usefulness, social influence, facilitating conditions, perceived security, technical support, and compatibility. The objective is to uncover underlying relationships and groupings among these variables to better understand their impact on adoption.

TABLE 3: FACTORS INFLUENCING ADOPTION

Factor Loadings	1	2	3
Perceived Ease of Use	-0.0259	-0.071091	-0.023457
Perceived Usefulness	-0.0427	0.986660	-0.123304
Social Influence	0.1480	0.065481	0.625152
Facilitating Conditions	-0.0444	0.013980	0.173343
Perceived Security	0.0200	-0.067307	0.007109
Technical Support	0.0296	0.047800	-0.149576
Compatibility	1.0006	-0.104033	-0.116390

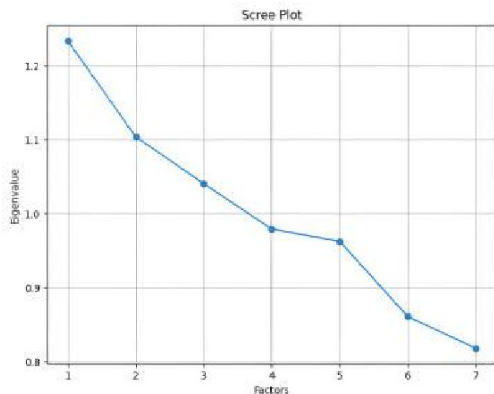


Fig.3 Scree plot

The exploratory factor analysis (EFA) results in Table 3 reveal distinct groupings among the variables influencing the adoption of digital payment platforms. Perceived usefulness strongly loads on Factor 2 (0.987), suggesting it is a critical driver in the adoption process. Social influence shows a notable loading on Factor 3 (0.625), highlighting its importance in influencing user behaviour. Compatibility has a significant loading on Factor 1 (1.001), indicating it is a primary consideration for adoption. Other variables such as perceived ease of use, facilitating conditions, perceived security, and technical support have lower loadings, suggesting they are less influential in this context. These findings help clarify which factors are most impactful in driving adoption.

D. Performance Metrics of Machine Learning Models

The effectiveness of the SVM model in predicting digital platform adoption using accuracy, precision, recall, and F1 score based on features such as perceived ease of use, perceived usefulness, social influence, facilitating conditions, perceived security, technical support, and compatibility. The objective is to assess the model's predictive performance for the target variable Digital Platforms.

TABLE 4: PERFORMANCE METRICS OF MACHINE LEARNING MODELS

Sl. No	Metric	Value
1	Accuracy	0.566667
2	Precision	0.593751
3	Recall	0.666667
4	F1 Score	0.662471

The performance metrics in Table 4 indicate the effectiveness of the SVM model in predicting digital platform adoption. The model achieves an accuracy of 56.67%, suggesting a moderate level of correctness in its predictions. The precision of 59.38% indicates that when the model predicts a specific digital platform, it is correct about 59.38% of the time. The recall of 66.67% shows that the model successfully identifies 66.67% of the actual digital platform adoptions. The F1 score of 66.25% balances precision and recall, reflecting a decent overall performance of the SVM model in predicting digital platform adoption based on the given features.

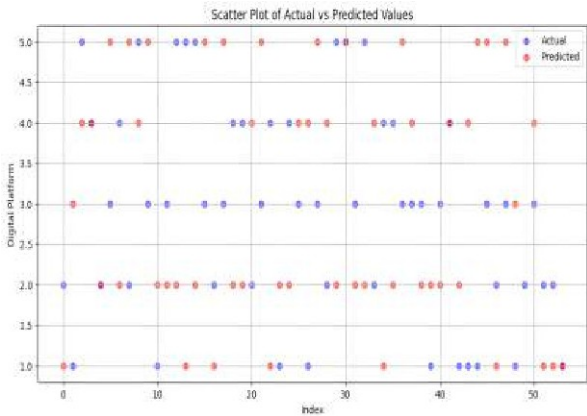


Fig.4 Scatter plot

E. Performance Analysis of Digital Platforms Based on Satisfaction

A comparative analysis of average satisfaction levels across various digital platforms to determine their relative performance. Objective to determine and visually compare the average satisfaction levels among different digital platfr

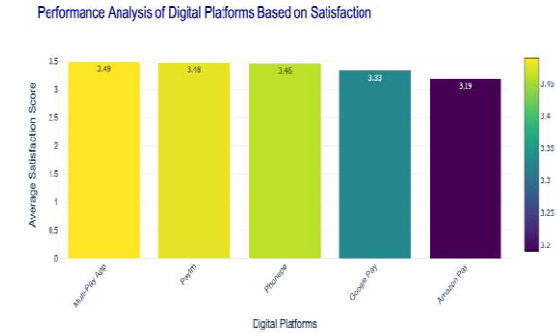


Fig.5 Performance analysis

The analysis of average satisfaction levels reveals that "Multi-Pay App" leads with the highest satisfaction score of 3.49, indicating it is the most positively perceived platform among the surveyed users. "Paytm" follows closely with a score of 3.48, suggesting strong user approval as well. "PhonePe" has a slightly lower satisfaction score of 3.46, while "GooglePay" and "AmazonPay" show lower satisfaction levels at 3.33 and 3.19, respectively. These results highlight that while most digital platforms are relatively well-regarded, there is a noticeable variance in user satisfaction, with "AmazonPay" trailing behind the others.

F. Predicting Digital Payments trends through Google trends

This analysis aims to forecast digital payment trends over the next five years using historical Google Trends data from the past five years. By applying the ARIMA model to quarterly data, we can identify and project future trends in digital payment interest, providing valuable insights for anticipating market dynamics and strategic planning in the fintech sector.



Fig.5 Google trends

The forecasted values from July 21, 2024, to May 23, 2027, consistently show an estimated value of approximately 37.30, with a gradually increasing upper

confidence interval (CI) and a narrowing lower CI. The forecasted value shall refrain analyzed values stay around 37.30, as the lower CI ascend from around 20.24 until approximately -32.28 on May 23, 2027. While the upper CI starts at 54.16 and reach up to approximately 106.87 over the same timeline. General trend indicates increasing uncertainty with the potential values discussion linearly.

IV. FINDINGS AND SUGGESTIONS

According to the study, the most essential factor with retail vendors use of digital payment platforms is perceived usefulness. When vendors believe a platform will provide value improving their operations or better service their customers, they are more likely to then use the platform. Social influence is another important element of use. Vendors are influenced by their peers, competitors, and consumers who use and advocate for digital payment platforms. This not only aligns with the UTAUT notion [2], [8], [21] but as critical consideration to itself also happens to be compatibility. Compatibility or woods for existing systems and or processes of the business. Vendors desire technologies that integrate effortlessly with their existing operations without requiring large adjustments. Contrary to predictions, perceived ease of use and technical support had little influence on adoption decisions[6], [10], [17]. This indicates that vendors are prepared to overcome usability issues if the perceived benefits are significant. While perceived security is a concern, it did not emerge as a significant driver of adoption. This could be attributed to growing confidence in the security procedures of established digital payment companies.

The examination of digital payment platform usage trends found that the number of transactions per day and transaction amounts have no significant impact on the classification of business categories. This suggests that the adoption of digital payment platforms is independent of the volume of daily transactions. The Support Vector Machine (SVM) model used to forecast digital payment platform adoption performed rather well, with an accuracy of 56.67%, precision of 59.38%, recall of 66.67%, and F1 score of 66.25%. This implies that, while the model is quite successful, its forecast accuracy might be improved [22], [23].

A comparison of satisfaction levels across various digital payment platforms indicated that "Multi-Pay App" and "Paytm" had the highest satisfaction rates, followed by "PhonePe". "GooglePay" and "AmazonPay" had lower satisfaction scores, indicating that these systems should improve user experience and solve specific vendor problems [7]. The ARIMA model predicts a constant central estimate for digital payment trends in the coming years, with an expanding range of uncertainty [3]. This indicates a consistent interest in digital payment platforms, however external variables may create substantial variations.

Digital payment platform providers should emphasize the actual advantages of their services, such as faster transaction times, greater customer support, and connection with other business tools. Including case studies and testimonials from other vendors can assist to underline these benefits [4], [6], [21]. Marketing strategy should focus on social proof and peer endorsements. Platforms could set up community

forums or user groups where suppliers can discuss their positive experiences and best practices. It is critical to ensure that digital payment platforms work with a variety of existing company systems and procedures. Providers should provide flexible integration options and support for a variety of business models [3], [9]. While these were not the key motivators, enhancing usability and providing strong technical assistance can increase overall satisfaction and lessen resistance to adoption. Simplifying user interfaces and providing specialized support channels may be helpful [8], [23]. Even if security was not a significant worry, continuing to engage in advanced security measures and publicizing these efforts to vendors helps foster trust and alleviate any remaining concerns. Providers with low satisfaction ratings, such as "GooglePay" and "AmazonPay," should study the specific challenges encountered by suppliers and apply focused improvements. Conducting regular satisfaction surveys and responding to input can help in this regard.

V. CONCLUSION

This study throws insight into the factors that influence the adoption of digital payment platforms by retail vendors in Bangalore. By combining conventional statistical approaches with modern machine learning techniques, the study reveals perceived usefulness, social influence, and compatibility as the primary adoption factors. The study also focuses on the SVM model's average predictive accuracy and the variable levels of satisfaction with different digital payment platforms. The findings highlight the necessity of meeting vendor-specific demands and preferences to increase adoption rates. Digital payment platform providers should prioritize demonstrating the practical benefits of their services, capturing social influence, assuring compatibility with existing systems, and resolving usability and security concerns. This allows for a more seamless move to digital payments while also improving retail vendors' overall efficiency and competitiveness.

VI. SCOPE OF FUTURE RESEARCH

Future research could investigate using different machine learning algorithms, such as neural networks or ensemble approaches, to improve the accuracy of adoption predictions. Combining multiple models may yield more robust and solid projected outcomes. Longitudinal studies that examine changes in adoption trends over time would provide a better understanding of how factors influencing adoption change. Comparing adoption patterns across different geographical areas may highlight regional changes and distinct issues. Qualitative research approaches, such as interviews and focus groups, may provide more in-depth insights about vendors' motives and barriers.

By addressing these issues, future research can build on the study's findings and contribute to a more complete knowledge of digital payment adoption among retailers. This, in turn, can inform policies and tactics to encourage widespread adoption of digital payments, benefiting both the retail sector and the whole economy.

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